

Connecting your virtual devices with real network hardware in GNS3

This can be achieved by two different methods. First is by binding every virtual router interface that needs to be connected to a switch to a dedicated physical interface on your host. Then, each dedicated interface can be connected to a real switch port.

Alternatively, you can create an 802.1Q trunk between your virtual network and a physical 'breakout switch.' Each virtual router interface is then assigned to a VLAN with tagging on the host side and the tagged traffic is received by the breakout switch via the single physical trunk link. VLANs are then distributed accordingly to real switches.



Fig 3: Routers

List of hardware which can be emulated by GNS3

- Cisco 1700 series
- Cisco 2600 series
- Cisco 3600 series
- Cisco 3700 series
- Cisco 7200 series
- Cisco PIX firewalls
- Cisco ASA firewalls
- Cisco IDS sensors
- Juniper routers

thor, its GUI is made with the powerful Qt library. SVG technology (scalable vector graphics) in its GUI provides high-quality symbols for designing your network topologies.

The user-friendly GUI window is divided into four panes and provides you with all the options to create network topologies at the very first sight. Different types of network nodes are available at the leftmost pane. Clicking onto any node symbol will provide you with the list of various types of that node. For example, different routers, Ethernet switch, ATM bridge, ATM switch, frame relay switch, cloud, Qemu and VirtualBox guests will appear after clicking on the specific node symbol.

The rightmost pane provides a topology summary where your connections can be viewed and nodes can be renamed. The top pane in the middle is your work area where you can build your topology graphically while the bottom pane is the GNS3 management console. A shortcut menu is at the top which provides all the relevant shortcuts for your working with the created topologies.

Memory and CPU usage

Running an IOS can consume up to 100 per cent of your CPU time and cause your computer to become very sluggish and prevent building more complex topologies. Also, building labs can consume a large amount of real and virtual memory.

GNS3 has in-built functionality to address these issues and ensure smooth functioning of the tool. An idle-pc value, dependent on the IOS version and not the PC, can be included to address the CPU usage. Ghostios and Sparmsem options are added to address the memory issues.

The idlepc value command performs analysis on a running image to determine the most likely points in the code that represent an idle loop in the IOS process and makes the virtual router to 'sleep' occasionally when this idle loop is executed, reducing the CPU consumption.

You can choose Idle PC by right clicking onto the router and an Idle PC value will be calculated, and then click the drop down arrow to see a list of possible idlepc values. You can select any one of the values with an asterisk (recommended values).

The Ghostios option significantly

reduces the amount of real host RAM needed for labs with multiple routers running the same IOS image, and the Sparmsem feature, on the other hand, does not conserve real memory, but reduces the amount of virtual memory used by your router instances.

You can select Preferences under the Edit menu and check the appropriate boxes to enable these features.

Other features

As mentioned above, GNS3 is not just a simulator, it is also an emulator. It can emulate a long list of router platforms and other devices. Additionally, if you use an EtherSwitch card in a router, switching platforms can also be emulated depending upon the degree of the card's supported functionality.

It uses a TCP/IP communications channel for the Dynamips hypervisor mode, which means you can run GNS3 on one machine and the Dynamips emulator on a different machine. So, GNS3 provides you the flexibility of working with a remote host machine, if required.

Another key feature of this tool is that you can use it to connect your virtual devices with real network hardware. Although the provided switching environment is limited for networking personnel who want to make more advanced switching labs, it is good enough for the people who want to study for their networking certifications.

GNS3 as a learning tool

GNS3 can emulate most of the router platforms and other network devices. You can use GNS3 as an alternative or complementary tool to real labs and simulate Cisco and Juniper topologies.

Also, the VirtualBox support in GNS3 allows even system administrators and engineers to utilise GNS3 to make labs and test network features.

Additionally, the high level of accuracy in simulations makes GNS3